



DEEP WORK

Rules for Focused Success
in a Distracted World

CAL NEWPORT

The Big “So What”



Deep work is about focusing without distraction on cognitively-demanding tasks. It’s vital for mastering hard things and delivering high-value outcomes. Yet, in our high-tech, hyper-connected society, people are constantly distracted by emails, social media and infotainment, losing their capacity for intense focus. **This book explains the importance of deep work and how to nurture this valuable but increasingly-rare skill.**

Introduction



Deep work is performed in a state of distraction-free concentration that pushes your skills and abilities to the limit. Such work creates real value and is hard to replicate. On the other hand, shallow work is cognitively undemanding and is often performed in a state of distraction. Such work doesn’t typically create much value and is easily replicated.



Many famous works are the result of deep work. Mark Twain wrote the bulk of “The Adventures of Tom Sawyer” in a secluded shed. Screenwriter and director Woody Allen did all his writing manually on a typewriter to stay free from digital distractions. In our modern economy, deep work is vital for high-impact activities ranging from writing a great novel to formulating a winning strategy.



Yet, technological and societal changes are pushing us increasingly toward a shallow work culture. A 2012 McKinsey study found that the average knowledge worker spent >60% of work hours on electronic communication and surfing the Internet. Such digital distractions are damaging as they (i) make us less effective in work that requires stretches of intense concentration and (ii) can permanently reduce our capacity to focus.

KEY QUOTES

“Deep work requires long periods of uninterrupted thinking.”

“Spend enough time in a state of frenetic shallowness and you permanently reduce your capacity to perform deep work.”

In this summary, we've distilled and reorganized the key ideas about Deep Work into 2 parts: "Why deep work is important" and "Rules and strategies for deep work".

Why Deep Work is Important



Deep work has value

• In the book "Race Against the Machine", economists Brynjolfsson and McAfee say that intelligent machines are rapidly changing our economy. They theorize that **in this new economy, people whose skills can be easily automated or outsourced will lose out, while 3 groups of people will win:**

- Highly-skilled workers (e.g. computer programmers or data analysts) who can work with machines to create new value.
- Superstars i.e. the top talents in each field. Technology has made it possible for employers to hire the best talents globally to work on projects remotely for a fixed fee. This can be more attractive than hiring full-time staff with average skills.
- Owners with the capital to invest in new technologies. Instagram, for instance, had only 13 staff but was sold for a billion dollars.



To become a highly-skilled worker or a superstar, you need **2 key abilities**. Deep work provides the foundation for both skills:

- Learn and master hard things quickly. To master complex machines and keep up with rapid technological changes, you must repeatedly learn hard things quickly. This requires intense concentration or deep work. Anders Ericsson discovered that "deliberate practice" was a key differentiator between normal vs outstanding performers—those who achieved top results did so because they (i) applied focused attention to improve a specific skill and (ii) received feedback to improve that skill precisely. This can be explained by neuroscience research: when you focus intensely on a specific skill, it forces an isolated neural circuit in the brain to fire repeatedly. This in turn triggers the production of myelin (fatty tissues) around those neurons to help them fire better (thus reinforcing the skill).

KEY QUOTES

"Once the talent market is made universally accessible, those at the peak of the market thrive while the rest suffer."

"Talent is not a commodity you can buy in bulk and combine to reach the needed levels: There's a premium to being the best."

"If you're comfortable going deep, you'll be comfortable mastering the increasingly complex systems and skills needed to thrive in our economy."

- Deliver top-notch results. Superstars don't just master valuable skills; they also apply them in a way that creates massive value, e.g. using computer programming to develop a solution to a real-world problem.
- o Jason Benn, originally an economics graduate, managed to learn a new skill and switch industry by going deep for 6 months. To force himself to learn computer programming, Benn locked himself in a room with just books and writing material. After 2 months of intense self-study, he attended a coding bootcamp, excelled there and got himself a job as a developer for double his previous salary.
- o When you switch between tasks, the "attention residue" from the previous task hinders your ability to fully focus and perform on the current task. By working singularly on 1 hard task for a long time, you minimize attention residue to drastically improve your effectiveness.



Deep work is uncommon

Deep work is valuable and should be a business priority. Yet, it is becoming increasingly uncommon at the workplace. This means there's a huge opportunity for individuals: **by mastering the rare skills and abilities for deep work, you can significantly increase your value in the new economy.**



There are at least 3 business trends that hinder deep work:

- Serendipitous collaboration. Serendipity is about putting random people together to spark unexpected ideas and synergies. This has led to open-office concepts which bring free interaction but also constant distraction.
- Instant communication. Instant messaging, phone messages and emails are now widely used in organizations, resulting in constant disruption and an overwhelming volume of random inputs.
- Having a constant social media presence. Social media is now seen to be a key marketing tool and companies may encourage their staff to post regular updates—such low-value work creates a distraction to other high-impact work.

KEY QUOTES

"To produce at your peak level you need to work for extended periods with full concentration on a single task free from distraction."

"Big trends in business today actively decrease people's ability to perform deep work, even though the benefits promised by these trends... are arguably dwarfed by the benefits that flow from a commitment to deep work."

 It's hard to reverse such trends due to 3 biases for shallow work:

- It's generally hard to quantify the impact of deep work vs shallow work. Without concrete stats, the Principle of Least Resistance kicks in and we do whatever is the easiest or most convenient. Yet, shallow work is costly in the long term. The CTO of Atlantic Media analysed their company-wide email usage in 2012 and found that every email was costing the company 95¢ in labor costs (or \$1mil per annum). In another experiment, a Boston Consulting Group team found that allowing its team member to take 1 day off work each week instead of being available 24x7 for clients actually resulted in better team outcomes.
- Most companies also lack clear productivity metrics for knowledge workers. Without clear metrics, people use busyness as a gauge for productivity. Doing more things may mean higher productivity for industrial workers, but it actually decreases the value of the output for knowledge workers.
- We're typically biased toward technology, i.e. we tend to assume that anything high-tech is generally good. Thus, we tend to adopt new technologies before stopping to debate the trade-offs.



Deep work improves your quality of life

Deep work can also improve your levels of happiness and fulfilment.



Neurological perspective: When you focus on something regularly, it strengthens the associated neural pathways in your brain. Over time, you can train your brain to habitually ignore the negatives and savor the positives to feel happier overall. By engaging in deep work regularly, you shut out trivial concerns and train your mind to see your work as rich and fulfilling.



Psychological perspective: The very experience of going deep into a challenge can be very fulfilling. Renowned psychologist Mihaly Csikszentmihalyi found that people felt best when they were in "flow"—when their minds or bodies were stretched to the limit in a quest to achieve something both difficult and worthwhile. In fact, his studies found that people often enjoyed work more than free time, since work allowed them to be immersed in meaningful goals, feedback and challenges.

KEY QUOTES

"Without clear metrics to support it, any business behavior is vulnerable to unstable whim and shifting forces."

"Knowledge work is not an assembly line, and extracting value from information is an activity that's often at odds with busyness, not supported by it."

"Clarity about what matters provides clarity about what does not."

"A deep life is a good life, any way you look at it."

"What we choose to focus on and what we choose to ignore—plays in defining the quality of our life."

"Human beings, it seems, are at their best when immersed deeply in something challenging."



Philosophical perspective: You can find meaning in any type of work regardless of the job description. The key is to adopt a craftsman's approach—don't treat it like a job, but a craft that you're seeking to master. Seek to hone your skills, deepen your appreciation and perfect your craft in your own unique way. Even a computer programmer can see well-written, simple/elegant code as something beautiful or poetic.

Rules and Strategies for Deep Work



In this section, we present 4 rules for deep work and their associated strategies.



Build rituals and routines

Even if you want to work deeply, it's not easy to do so because most office setups and work cultures are not ideal for deep work. You'd also have to overcome your natural instinct to switch to easy, distracting activities (e.g. checking emails or surfing social media sites). **Build rituals and routines so it's easier to overcome obstacles and move into states of intense focus.**



Tailor your depth approach to match your circumstances.

- **Monastic approach:** completely remove or cut down shallow work. For example, computer scientist Donald Knuth deleted his email accounts and returned to communication via postal mails, with an admin assistant to sort through his letters and bring only the important ones to his attention. This approach works best if you have a clear, concrete contribution that's pegged to your individual effort.
- **Bimodal approach:** Split your time so you devote stretches of time for deep work and leave the rest for other obligations. To achieve extreme productivity, set aside minimally 1 full day each time for deep work. Try blocking out a few days per week or an entire season per year just for deep work. For example, psychologist Carl Jung took regular retreats to Bollingen to think and write, but immersed himself back into his usual meetings, clinical practice and digital connections once he returned to Zurich. This model is compatible with most jobs, though it may take some time to adjust.

KEY QUOTES

"You don't need a rarified job; you need instead a rarified approach to your work."

- Rhythmic approach: Create a rhythm for deep work daily so it becomes a habit. You can use visual indicators (e.g. marking a big red X on your calendar each time you finish a stretch of deep work or a deep output), or set a distinct time period to start and end deep work every day. This approach won't have the same impact as going deep for 1 full day, but it's useful if your self-discipline or work schedule doesn't allow you to devote full days for deep work.
- Journalistic approach: Fit deep work into your schedule whenever you can. This approach is not for novices as task-switching is usually counter-productive. Walter Isaacson could write his first book using this approach because he already had years of journalism experience and could switch into writing mode at will.



Build strict rituals so you can transit smoothly into your deep work sessions without having to decide on the elements each time.

- Pulitzer Prize winner Robert Caro trained himself to be organized by creating strict rules, from where he places his books to what he wears to the office. When Charles Darwin was working on "On the Origin of Species", he followed a strict daily routine: wake up at 7am, take a walk, eat breakfast alone, work for 1 hour in his study, read his letters for 1 hour, then continue for 1.5 hours in his study till noon.
- Consider these factors when tailoring your ideal ritual:
 - (i) Location and duration. Ideally, find a place dedicated to deep work and set a duration that makes each session challenging.
 - (ii) Rules for how you'll work, e.g. specific processes, no internet.
 - (iii) Supporting elements, e.g. having a healthy meal and a cup of coffee before you start.



Make grand gestures that break your routine and involve a big investment of time, money or effort, e.g. travel to an exotic location or lock yourself up to finish a project.

KEY QUOTES

"By supporting deep work with rock-solid routines that make sure a little bit gets done on a regular basis, the rhythmic scheduler will often log a larger total number of deep hours per year."

*"(Great creative minds) think like artists but work like accountants."
- David Brooks*

"To make the most out of your deep work sessions, build rituals of...strictness and idiosyncrasy."

"Sometimes to go deep, you must first go big."

- Microsoft founder Bill Gates retreats into the forest each year for 2 weeks to think and read. Writers like Daniel Pink and Michael Pollan built writing cabins for focused writing. Entrepreneur Peter Shankman invested 30 hours and \$4,000 for a round-trip business-class ticket just so he could focus on finishing a book manuscript in the plane/lounge.
- Such grand gestures provide a refreshing change of environment and commit you psychologically to get things done.



Collaborate. Deep work need not be done in isolation and can be enhanced through collaboration. The key is to separate serendipitous collaboration from deep work sessions.

- It's ideal to have a hub-and-spoke office setup with both common spaces (e.g. shared hallways for people to connect) and private corners/offices (for people to think/work quietly).
- You can hold deep work sessions by jointly working on a problem with someone to stimulate ideas and push deeper into the issue.



Execute effectively. Use the framework from “The 4 Disciplines of Execution” to roll out your plan seamlessly:

- Discipline 1: Focus on just a few “wildly important goals” (WIGs). To work deeply, don't try to avoid distractions. Instead, focus on a goal that excites you enough to shut out those distractions.
- Discipline 2: Act on the lead measures, i.e. the activities that drive your end-results. The lead measure for deep work is the amount of time spent working deeply toward your WIGs.
- Discipline 3: Have a compelling scoreboard so you can keep score. Newport created a visible scoreboard where he (i) tallied the # hours spent weekly on deep work and (ii) circled out the tally whenever he completed a big milestone. This helped him to connect deep work hours with results, and gave him a realistic idea of how much deep work he needed for each milestone.
- Discipline 4: Create a rhythm of accountability. Be accountable to yourself through weekly reviews: examine the deep work you've done the previous week, celebrate good results, identify learning points, and set new commitments the following week.

KEY QUOTES

“When it comes to deep work... consider the use of collaboration when appropriate, as it can push your results to a new level.”

“It's often straightforward to identify a strategy needed to achieve a goal, but what trips up companies is figuring out how to execute the strategy once identified.”



Go for downtime. At the end of your work day, learn to put away all work issues physically and mentally.

- Idle time is not wasteful; it's actually essential for optimal performance. Switching off from work regularly improves your results because (i) it allows your subconscious mind to work on complex problems that your conscious mind can't unravel and (ii) it allows you to recharge so you're more productive at your next deep work session. In any case, the type of shallow tasks we fit into the evenings typically create little value.
- Develop a "shutdown ritual" to help you put away your work totally: check your emails to ensure there're no urgent follow-ups required, list down all the tasks on your mind so you can find them the next day, check your calendar to ensure there're no vital events you've missed, have a rough plan of action for the next day, and then say to yourself "shutdown complete".



Train your mind

Our mental muscles grow strong with use and weak with disuse. People who constantly multi-task have been found to become chronically distracted, i.e. they can't concentrate even when they want to. Learn to resist distractions and improve your capacity for intense focus.



Make focus your default state of mind. Don't schedule periods of time away from distractions; schedule time for the distractions instead.

- Often, people will start looking at their phones the moment they enter a queue or feel bored—this trains the mind to constantly seek novelty. Such habits can't be weaned off with occasional distraction-free periods, e.g. an internet-free day. You must operate distraction-free by default.
- Fix specific time blocks for distracting activities like TV or internet and avoid them totally outside of those time blocks. Do this both at work and at home. If your work requires you to be connected, then create more internet blocks but stick to them strictly.



Compress your timelines. Think of a complex/hard task, estimate the amount of time you'd typically need, then give yourself a drastically reduced timeline to complete it. Ideally, make this goal

KEY QUOTES

"Idleness is not just a vacation, an indulgence or a vice; it is as indispensable to the brain as vitamin D is to the body...it is, paradoxically, necessary to getting any work done."
- Tim Kreider

"When you work, work hard. When you're done, be done."

"The ability to concentrate intensely is a skill that must be trained."

"Instead of scheduling the occasional break from distraction so you can focus, you should instead schedule the occasional break from focus to give in to distraction."

public. This will force you to tackle the task with intense focus. Do this for once a week initially then gradually increase the frequency as your mind adapts to the intensity of the sessions. After a few months, you should see a marked improvement in your focus and performance.



Meditate productively. Find periods of time when you're occupied physically but not mentally (e.g. when you're walking or showering). Use that time to think deeply about a concrete professional problem, e.g. a presentation outline or a business strategy. Aim for at least 2-3 such sessions a week.

- Catch yourself if you start looping over what you already know instead of doing the hard work of thinking. Each time your mind wanders, bring it back to the problem at hand.
- Develop a structure for your deep thinking. For example, you can define the variables, the next-step question (e.g. "How can I deliver this presentation powerfully?") and review your answer.



Do memory training. Memory athletes can achieve cognitive feats because they've trained themselves to develop a superb attention span. Use this technique by Ron White to memorize a deck of cards:

- Imagine yourself walking through 5 rooms in your house. Place 10 items (e.g. a welcome mat, a pen) in each room and add 2 items in the backyard. Repeat the process until you can recall all 52 items easily.
- Associate each of the 52 playing cards with a memorable person or thing, e.g. Queen of Hearts with Queen of England.
- Shuffle the cards. As you draw each card from the deck, place the associated person/thing sequentially in the house, e.g. the Queen of England wiping her shoes on the welcome mat. Memorize the entire deck of cards using this visual technique.



Use digital media selectively

Network tools can be useful for your success and happiness, yet most of them are counter-productive. Instead of using any tool that offers a benefit. **Newport advocates selecting your tools carefully to use only tools that have a huge nett positive impact on your goals.**

KEY QUOTES

"Your ability to concentrate is only as strong as your commitment to train it."

"To master the art of deep work...you must take back control of your time and attention from the many diversions that attempt to steal them."

 **Focus on the vital few.** In most aspects of life, 80% of outcomes come from 20% of causes. While many things can give us minor benefits, only a handful will give us massive results.

- Apply the 80-20 rule to your internet habits. Identify your top personal and professional goals, and 2-3 most critical activities to help you achieve them. Evaluate each network tool against these goals/activities. Use only tools that have substantial positive impact that outweigh any negative impact.
- By dropping most tools with only marginal benefits, you free up the time and energy for a few top tools that'll deliver much greater payoffs. For example, if your goal is to build deep relationships with people who matter most to you, then it's more productive to catch up with a close friend over coffee than spend time on Facebook.

 **Use leisure time wisely.** You have total freedom to do whatever you want with the leisure time outside of work. Don't use it to mindlessly surf the internet. By planning your leisure time carefully and filling it with quality activities that enrich your mind, you'll end up feeling more energized and fulfilled.

 **Do a social media embargo.** Of all network tools, social media tends to be the most addictive and disruptive. Stop using all social media services for 30 days without announcing your intent or deactivating your accounts. Then, ask yourself: (i) Were the last 30 days noticeably worse without the service and (ii) Did people care that you weren't on the service? Unless the answer is "yes" to both questions, stop using that service.



Limit shallow work

It's not necessary nor possible to eliminate all shallow work. In any case, the average person only has enough cognitive capacity for about 4 hours of intense deep work a day. The key is to **take control of your shallow work, so they don't hijack your time/energy for deep work.**

 **Schedule each day in advance.** List down your key priorities and activities, then assign them to time blocks so every minute of your day is accounted for. You should also include time blocks for meals and relaxation.

- As the day unfolds, you may find that you've underestimated the time needed for certain tasks, or you may get interrupted

KEY QUOTES

"All activities, regardless of their importance, consume your same limited store of time and attention."

"Shallow work...doesn't become dangerous until after you add enough to begin to crowd out your bounded deep efforts for the day."

by unexpected events. You can (i) revise your schedule, (ii) create conditional time blocks to be used differently depending on how earlier events unfold or (iii) create time blocks for contingencies.

- The goal is not to follow the schedule rigidly but to get better at estimating your time requirements and to be deliberate about how you spend your time.



Quantify depth. Assess how deep every activity is by asking: “How many months would it take to train a smart, fresh college graduate to complete this task?” Tackle the activities progressively down the deep-to-shallow scale.



Set a quota for shallow work. Ask yourself: what percentage of my time should be spent on shallow work? For most non-entry level jobs, the range should be between 30-50%.

- Stick to the quota so you’d (i) reject some shallow work to free up time for higher-value deep work and (ii) become more selective with your shallow work.
- If you’re an employee, discuss this quota with your boss. If your boss doesn’t believe in the need for deep work, then it’s a sign that your job may not be ideal for long-term career growth.



Artificially limit your work hours. Instead of extending your work hours to fit the workload, fix your work schedule (e.g. don’t work beyond 5pm or 5.30pm). Work backward to fix the rules and habits needed to complete your goals within the available time. You’ll be forced to turn down unessential tasks and requests which you may otherwise have accepted. Newport calls this “fixed-schedule productivity”.



Take control of your emails. Don’t be overly accessible.

- **Make the sender work harder.** Don’t offer general-purpose emails on your website. Use a sender-filter: list down different email addresses for specific purposes and set expectations on if/when you’ll respond, e.g. “We welcome interesting proposals that meet the following criteria. We’ll only be responding to shortlisted proposals.” This frees you from the obligation to reply to every single message.

KEY QUOTES

“The shallow work that increasingly dominates the time and attention of knowledge workers is less vital than it often seems in

- Invest time to craft a concrete email. Resist the urge to send quick/vague replies or open-ended questions like “Thoughts?” Pause to consider the project involved and how you can bring it to a successful close with the minimal messages. Then, craft a process-centric email that describes the process you’ve identified, specifying clearly the next step(s) to take. Ideally, aim to close the loop in 1 email.
- As a rule of thumb, don’t reply to emails that (i) are ambiguous or hard to give a straightforward response; (ii) are irrelevant to you, or (iii) have no good/bad outcomes regardless of your reply.

Conclusion and Other Details in the Book



Deep work involves hard work and massive changes to your work habits. However, by going deep, you can drastically improve your levels of productivity and fulfillment to become much more valuable in the modern economy.

In the book, Newport shares many of his personal experiences, observations and other references to support his theories and recommendations. For more details, please visit calnewport.com/books/deep-work/.

About the Author

Dr. Calvin Newport (born 1982) is an author and associate professor of computer science at Georgetown University. He completed his undergraduate studies at Dartmouth College and received a Ph.D. from MIT in computer science. He spent 2 years as a post-doctoral associate at the MIT before joining Georgetown University as an assistant professor of computer science.



Dr. Newport also runs a blog, Study Hacks, which focuses on academic and career success. He’s an advocator of “deep work” and “digital minimalism”.

KEY QUOTES

“To leave the distracted masses to join the focused few...is a transformative experience.”